

Meta-Narrative Theory: Mathematical Foundations

Table of Contents

1. Duality of Emergence: Core Mathematical Formalism
2. Generative Axiom: Recursive Generativity
3. Stability Axiom: Persistence and Thresholds
4. Narrative-Math Mapping Axiom
5. Emergence Axiom: Novelty and Complexity
6. Quantum Extensions
7. Example Calculations and Diagrams

1. Duality of Emergence: Core Mathematical Formalism

Summary:

Emergence of structure, information, or complexity is only possible when at least two distinguishable states or entities exist in relation. This is the minimal mathematical condition for emergence.

Formalism:

$$|S| \geq 2, \quad R : S \times S \rightarrow P, \quad R(a, b) \neq R(a, a) \text{ for } a \neq b$$

Where S is a set of states, P is a set of properties or processes, and R is a relation generating new properties from distinction.

2. Generative Axiom: Recursive Generativity

Summary:

True generativity requires duality. New states emerge from the interaction of distinct elements, and recursion enables the growth of complexity.

Formalism:

$$G(S) = S \cup \{f(a, b)\} \quad \text{for } a, b \in S, a \neq b$$

- Initial: $S_0 = \emptyset$ (MoP: pure potential)
- First distinction: $S_1 = \{E, G\}$ (MoM: duality)
- Recursive: $S_{n+1} = S_n \cup \{f(a, b)\}$
- Multiplicative phase: $|S_{n+1}| = k \cdot |S_n|, k > 1$

3. Stability Axiom: Persistence and Thresholds

Summary:

A system is stable if it maintains sufficient duality, generativity, agency, and observation. Stability is a dynamic balance, not stasis.

Formalism:

$$S \text{ is stable} \iff D + G + A + O \geq T$$

Where:

- D : ongoing distinction/duality
- G : generativity
- A : agency
- O : observation
- T : stability threshold

4. Narrative-Math Mapping Axiom

Summary:

All generative narrative acts require duality and map to mathematical operations. This axiom formalizes the correspondence between narrative events and mathematical operations.

Formalism:

$$\forall N, \exists M : N \mapsto M \quad \text{if and only if duality exists}$$

Where N is a narrative act, M is its mathematical counterpart.

5. Emergence Axiom: Novelty and Complexity

Summary:

Emergence occurs only when duality is present and recursive generativity is possible. Novelty and complexity arise from ongoing synthesis and interaction.

Formalism:

$$D > 0 \wedge \exists P : P \notin S_0, P \in S_n \text{ for some } n > 0$$

Where P is a property not present in the initial state but appears after recursive generativity.

6. Quantum Extensions

Quantum Observation and Agency

Observation:

$$O_q(|\psi\rangle) \rightarrow |\phi\rangle$$

Observation collapses a superposed state $|\psi\rangle$ to an eigenstate $|\phi\rangle$.

Agency:

$A_q(|\psi\rangle) \rightarrow$ new superposition or entangled state

7. Example Calculations and Diagrams

Example: Generative Recursion

- Start: $S_0 = \emptyset$
- First distinction: $S_1 = \{E, G\}$
- Next: $S_2 = \{E, G, f(E, G)\}$
- Continue: $S_3 = S_2 \cup \{f(G, f(E, G))\}$

Stability Example

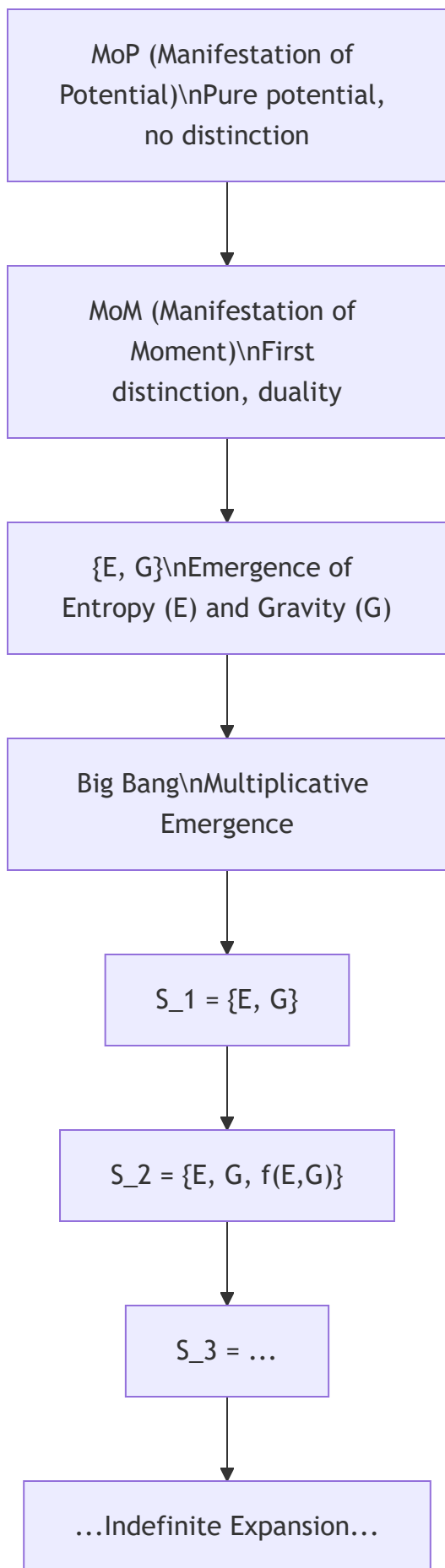
If $D = 1, G = 1, A = 1, O = 1, T = 3.5$, then S is stable because $1 + 1 + 1 + 1 = 4 \geq 3.5$.

Emergence Example

If S_0 is pure potential ($\emptyset$), and after the first distinction ($S_1 = \{E, G\}$), recursive generativity produces new emergent properties, then emergence has occurred.

Diagram: Emergence from MoP to Big Bang (Mermaid)





This document collects the core mathematical content of the meta-narrative theory for reference and export.